

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

***Artificial Intelligence
and Society***



Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 3.3: Data Transformations

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Motivation

- We have already seen an example of **feature engineering** for data transformation: one-hot encoding.
- We can do a lot more to make the data we have more useful for machine learning applications.
- On the other hand, data can also have **too many attributes** / features.
- Reducing the number of attributes is called **dimensionality reduction**.

Feature Engineering

Example: Headhunter

- We work at a headhunting agency and want to find out which candidates we should focus our efforts on.
- We use information on demographics, income, and education to predict employability.

AGE	INCOME	GENDER	PHONE NR.	DEGREE	PLACED
23	38.000	male	+43 664 1467812	bachelor	yes
35	52.000	female	+49 176 0378193	master	yes
42	93.000	female	+43 676 5672899	PhD	yes
31	39.000	male	+43 690 8848782		no
54	84.000	diverse	+49 170 2340055	master	yes
48	45.000	male	+43 664 1919822	bachelor	no

One-Hot Encoding and Binary Attributes

- We transform boolean values into ones and zeros.
- We use one-hot encoding to separate categorical attributes into multiple columns with binary attributes.

DEGREE_NONE	DEGREE_BACH	DEGREE_MAST	DEGREE_PHD	PLACED
0	1	0	0	1
0	0	1	0	1
0	0	0	1	1
1	0	0	0	0
0	0	1	0	1
0	1	0	0	0

Standardization

- Different value ranges or **feature scales** can lead to large numbers getting more weight than small ones, even if this is not intended.
- We **standardize** numerical attributes by subtracting the mean value of the attribute and deviding by the standard deviation (also called Z-score normalization).

AGE	INCOME	GENDER	PHONE NR.	DEGREE
-1.39	-0.86	male	+43 664 1467812	bachelor
-0.34	-0.27	female	+49 176 0378193	master
0.28	1.40	female	+43 676 5672899	PhD
-0.69	-0.81	male	+43 690 8848782	
1.33	1.07	diverse	+49 170 2340055	master
0.80	-0.56	male	+43 664 1919822	bachelor

Deleting Features

- Some attributes of people are **protected** and may not be used for decisionmaking.
- Gender is one of these attributes, as is for example religion and ethnicity – we remove it.

legal aspects
of AI

ethical aspects
of AI

AGE	INCOME		PHONE NR.	DEGREE
-1.39	-0.86		+43 664 1467812	bachelor
-0.34	-0.27		+49 176 0378193	master
0.28	1.40		+43 676 5672899	PhD
-0.69	-0.81		+43 690 8848782	
1.33	1.07		+49 170 2340055	master
0.80	-0.56		+43 664 1919822	bachelor

New Attributes

- From how much annual income a person earns per age, conclusions can perhaps be drawn about the person's ambitions.
- We **construct a new attribute** income / age.
- The phone numbers are useless as they are assigned randomly. But from the area code, we can draw conclusions about the country of residence.

AGE	INCOME	INCOME_AGE	PHONE NR.	DEGREE
-1.39	-0.86	0.62	Austria	bachelor
-0.34	-0.27	0.79	Germany	master
0.28	1.40	5.00	Austria	PhD
-0.69	-0.81	1.17	Austria	
1.33	1.07	0.80	Germany	master
0.80	-0.56	-0.70	Austria	bachelor

Dimensionality Reduction

When We Have Too Many Features ...

Problem 1: Large Data Volumes

Let's recall the example with the text:

- There are around 600,000 words in the Oxford English Dictionary.
- To represent a sentence with 6 words, we need a 600,000 x 6 cell table.
- This would occupy $600,000 \times 6 \times 32 / 8 / 1024 / 1024 = 13.7$ Megabytes of memory* $\rightarrow$ very inefficient.

*Even if we store the numbers not as 32-bit integers but only as 2-bit boolean values, it's still 0.86 MB!

	I	take	the	dog	outside
I	1	0	0	0	0
take	0	1	0	0	0
the	0	0	1	0	0
dog	0	0	0	1	0
outside	0	0	0	0	1
mouse	0	0	0	0	0
cat	0	0	0	0	0
house	0	0	0	0	0
	...	...	...	...	...
you	0	0	0	0	0

The Curse of Dimensionality

- For each additional feature the **space** in which the data resides gains another dimension. With each dimension, the **volume** increases rapidly.
- **Consequence 1: Overfitting**
100 observations cover the (one-dimensional) space between 0 and 1 well. In a 10-dimensional space, they are far too few to have enough similar examples for an algorithm to learn general patterns.
- **Consequence 2: Difficulty of finding clusters**
The difference between the smallest and largest **distance** between two data points becomes increasingly small.

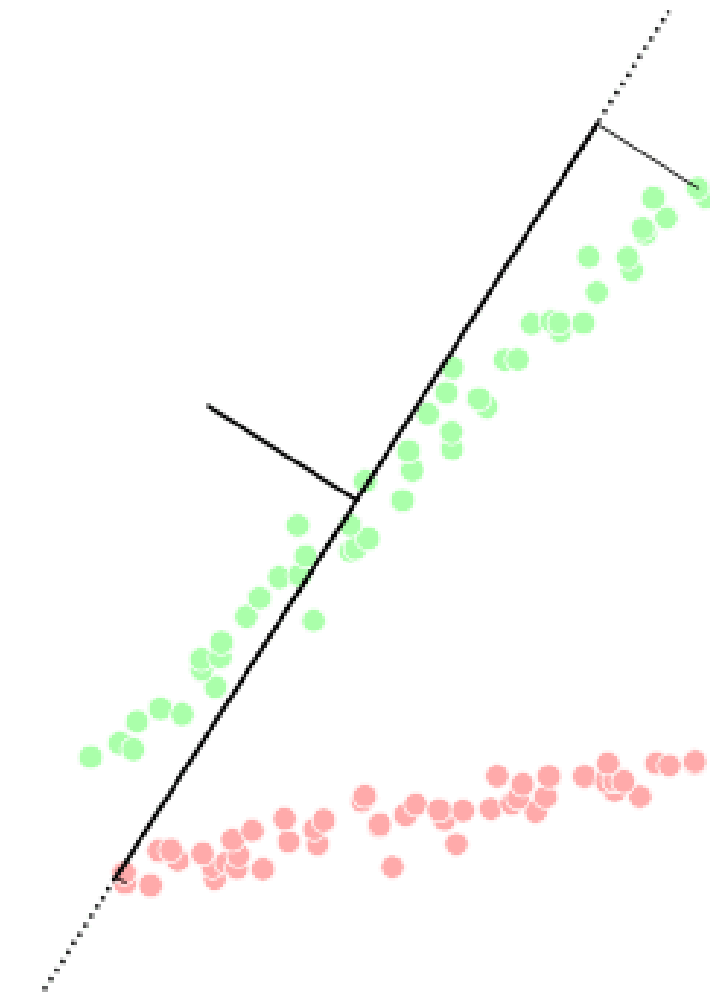
See this [post](#) for an explanation.

→ **Rule of thumb:** five observations are needed for every additional feature in the data set.

→ **Distance measures** like Euclidean distance don't work well for high-dimensional data. Use **similarity measures** (Cosine similarity) instead.

Solution: Dimensionality Reduction

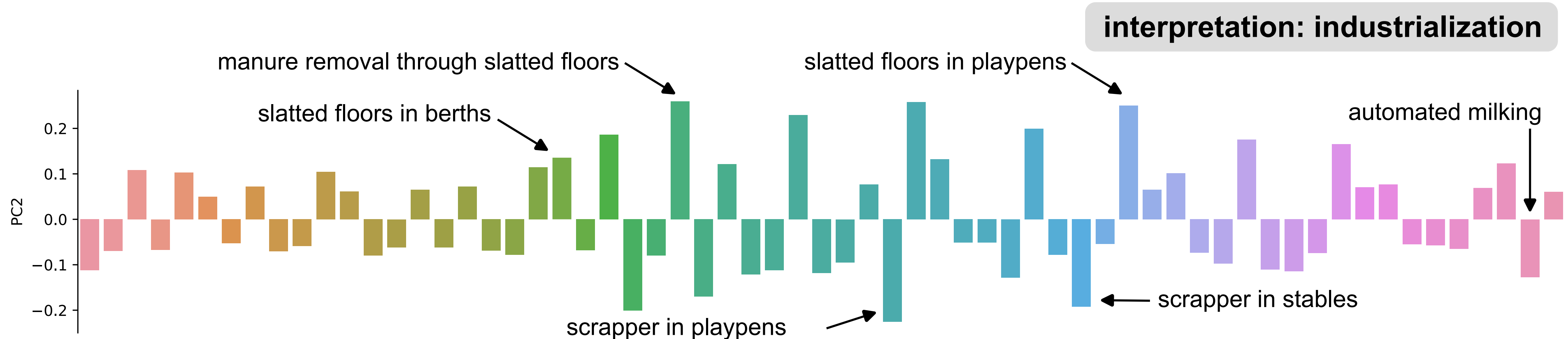
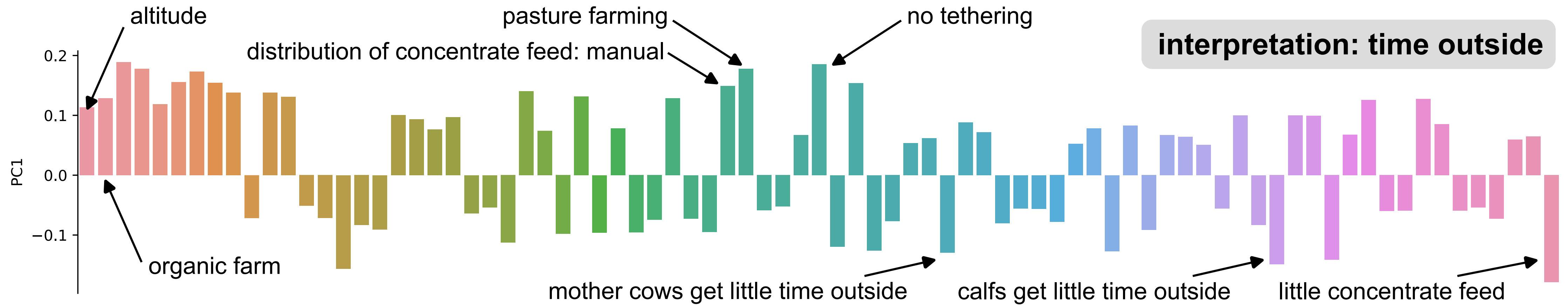
- **Feature Selection:** Remove features that do not carry important information.
- **Projection:** We search for a projection of the data points into a space where differences between data points are maximized.
- One method by which this can be achieved is **Principal Component Analysis (PCA)**.
- The result of a PCA is a projection of the data into a space where the **variance of the data points is concentrated in a few dimensions**.



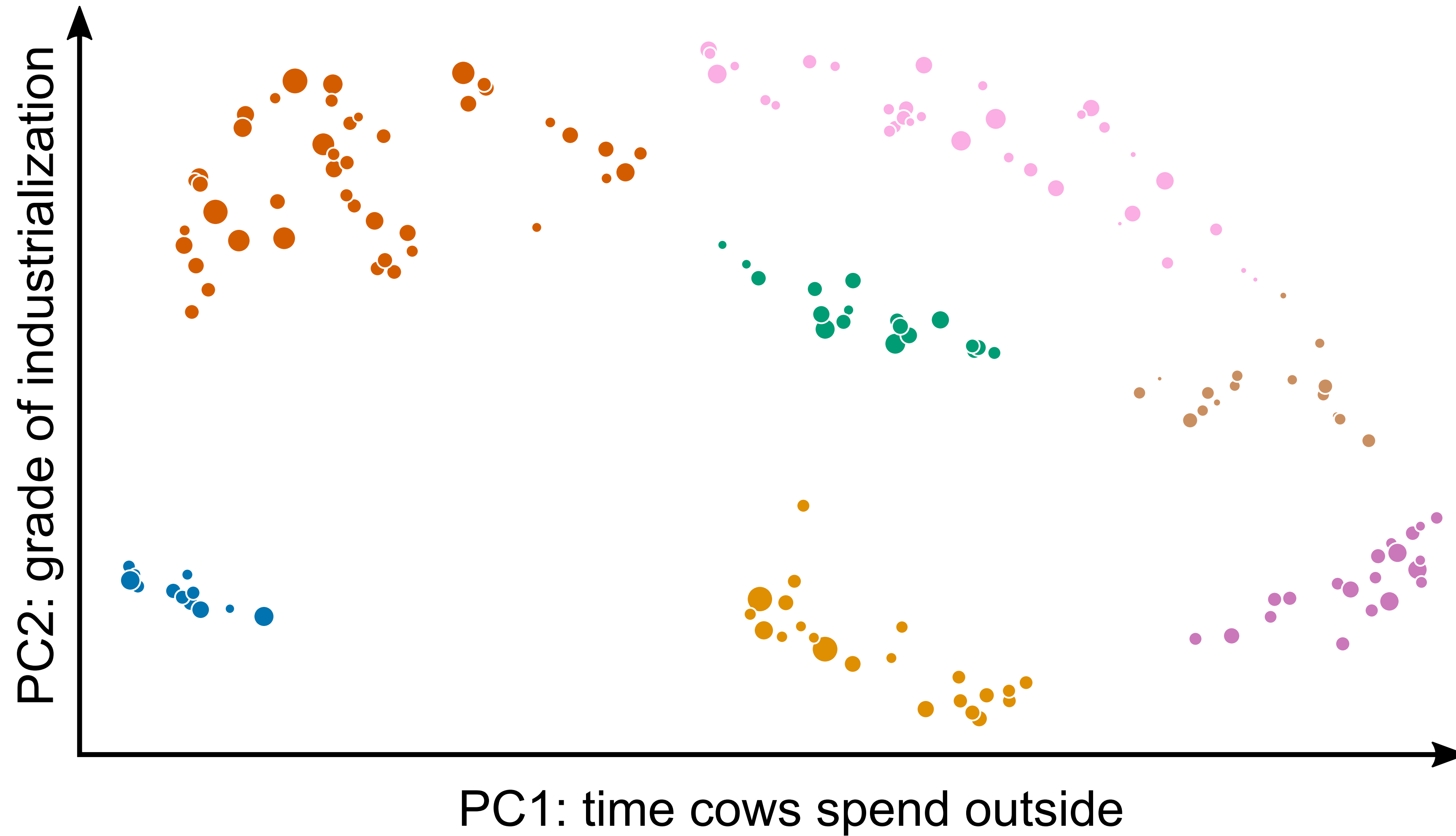
Visualization of a projection with PCA in two dimensions. Source: [Wikipedia](#), CC BY-SA 4.0.

→ The result of PCA initially has as many dimensions as the original space, but only a few are truly important. **We can remove the other dimensions** without losing much information.

Example PCA for Farms



Dairy farms in Austria



Note: Here we actually used a different dimensionality reduction technique ([UMAP](#)). The difference is that PCA is a linear combination of dimensions whereas UMAP allows for nonlinear combinations.

Summary

- **Feature Engineering** helps us get the most out of our data and avoid problems with machine learning algorithms or legal requirements. Techniques include:
 - One-hot encoding
 - Construction of new attributes
 - Standardization
 - Deletion of features
- **Dimensionality Reduction** helps us to avoid problems with too many features (Curse of Dimensionality). Problems include:
 - High memory and computation requirements
 - Overfitting
 - Difficulty of finding clusters when using distance measures
- Principal Component Analysis (PCA) is an approach to dimensionality reduction. It concentrates the variance contained in the data into a few dimensions, allowing us to delete the rest.